

Product Specification

Part Name: TFT LCD Display Module

Customer Part ID:

Cute Orange ID: KAC-N200-2432KHWIG20-A8

Ver: 1.0

Customer:

Approved by

From: Cute Orange technology Co.,Ltd.

Approved by

Notes:

1. Please contact Cute orange technology Co.,Ltd. before assigning your product based on this module specification.
2. The information contained herein is presented merely to indicate the characteristics and performance of our products. No responsibility is assumed by Cute orange technology Co.,Ltd. for any intellectual property claims or other problems that may result from application based on the module described herein.

2

Table of Contents

TFT LCD Display Module.....	1
1.General Description.....	4
2. Features.....	4
3. Mechanical Specification.....	4
4. Product picture.....	5
5. Pin Descriptions.....	5
6. Mechanical Dimension.....	6
7. Schematic diagram.....	6
8. Absolute Maximum Ratings.....	7
9. Electrical Characteristics.....	7
10. AC Characteristics.....	8
11.Command Table.....	8
12. Recommended Setting and Initialization Flow for Reference.....	8
13.Optical Specifications.....	9
13.1 Optical Specifications.....	9
13.2 Definition of Response Time.....	9
13.3 Definition of Contrast Ratio.....	10
13.4 Definition of Viewing Angles.....	11
13.5 Definition of Color Appearance.....	11
13.6 Definition of Surface Luminance, Uniformity and Transmittance.....	12
14 Quality Assurance.....	13
14.1 Purpose.....	13
14.2 Agreement Items.....	13
14.3 Standard of the Product Visual Inspection.....	13
14.4 Inspection Specification.....	14
14.5 Classification of Defects.....	18
14.6 Identification/marketing criteria.....	18
14.7 Packing.....	18
15 Reliability Specification.....	18
16 Precautions and Warranty.....	19
16.1 Safety.....	20
16.2 Handling.....	20
16.3 Operation.....	20
16.4 Static Electricity.....	20
16.5 Limited Warranty.....	20
17 Packaging.....	21
18 Prior Consult Matter.....	21

1 General Description

KAC-N200-2432KHWIG20-A8 This display module is a transmissive type color active matrix TFT(Thin Film Transistor) liquid crystal display (LCD) that uses amorphous silicon TFT as a switching device. This module is composed of a TFT LCD module, a driver circuit, and a back-light unit.The resolution of a 1.9" contains 170(RGB)X320 dots and can display up to 262k colors.

2 . Features

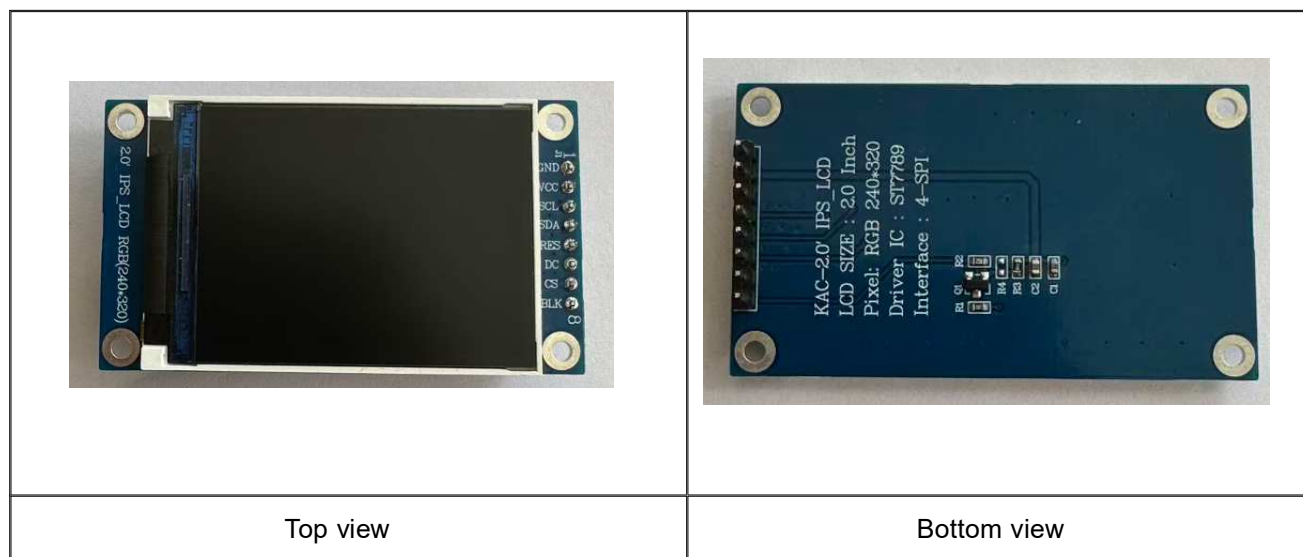
Note 1: Excluding hooks, posts , FPC/FPC tail etc.

NO.	Item	Contents	Unit
1	LCD Size	2.0	inch
2	Display Mode	Normally black	-
3	Resolution	240(H)RGB x320(V)	pixels
4	Pixel pitch	0.1275(H) x 0.1275(V)	mm
5	Active area	30.6(H) x 40.8(V) mm	mm
6	Module size	35.7(H) x 51.2(V) x2.2(D) mm	mm
7	Pixel arrangement	RGB Vertical stripe	-
8	Interface	4 Line SPI	-
9	Display Colors	65K	colors
10	Drive IC	ST7789V2	-
11	Luminance(cd/m2)	400 (TYP)	Cd/m2
12	Viewing Direction	All View	Best image
13	Backlight	4 White LED Parallel	-
14	Operating Temp.	-20℃~ + 70℃	℃
15	Storage Temp.	-30℃~+ 80℃	℃
16	Weight	7.9	g

3. Mechanical Specification

Item	Specifications	Unit
Dimensional outline	63.2(H)*29(V)*3.0MAX	mm
Resolution	240(H)RGB*320(V)	dots
Pixel size	0.1275(H) x 0.1275(V)	mm

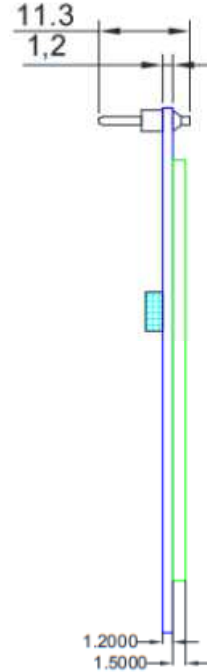
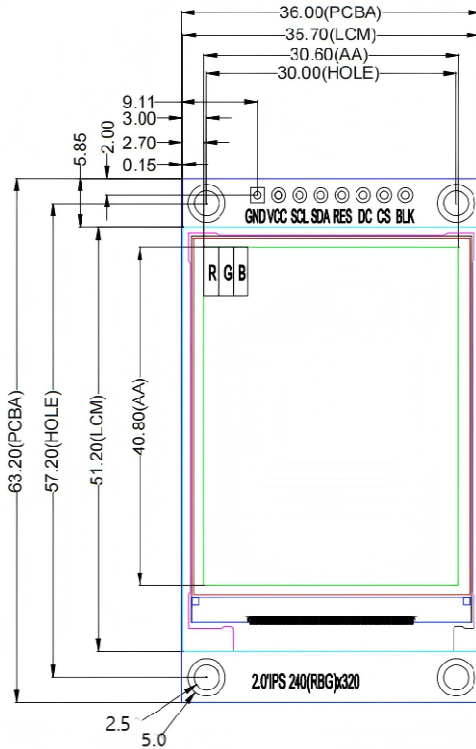
4. Product picture



5. Pin Descriptions

PIN No.	Symbol	Description
1	GND	Ground of Logic Circuit This is a ground pin. It acts as a reference for the logic pins. It must be connected to external ground
2	VCC	Power Supply for Logic This is a voltage supply pin. It must be connected to external source
3	SCL	The serial clock input
4	SDA	The serial data input
5	RES	Power Reset for Controller and Driver This pin is reset signal input. When the pin is low, initialization of the chip is executed. Keep this pin pull high during normal operation
6	DC	Data/Command Control This pin is Data/Command control pin. When the pin is pulled high, the input at SDA is treated as display data. When the pin is pulled low, the input at SDA will be transferred to the command register.
7	CS	Chip Select This pin is the chip select input. The chip is enabled for MCU communication only when CS# is pulled low.
8	BLK	Backlight control pin When the pin is pulled high turn on backlight, When the pin is pulled low turn off backlight

6.Mechanical Dimension



backlight circuit diagram



CIRCUIT DIAGRAM
(I=60MA V=2.9-3.3V)

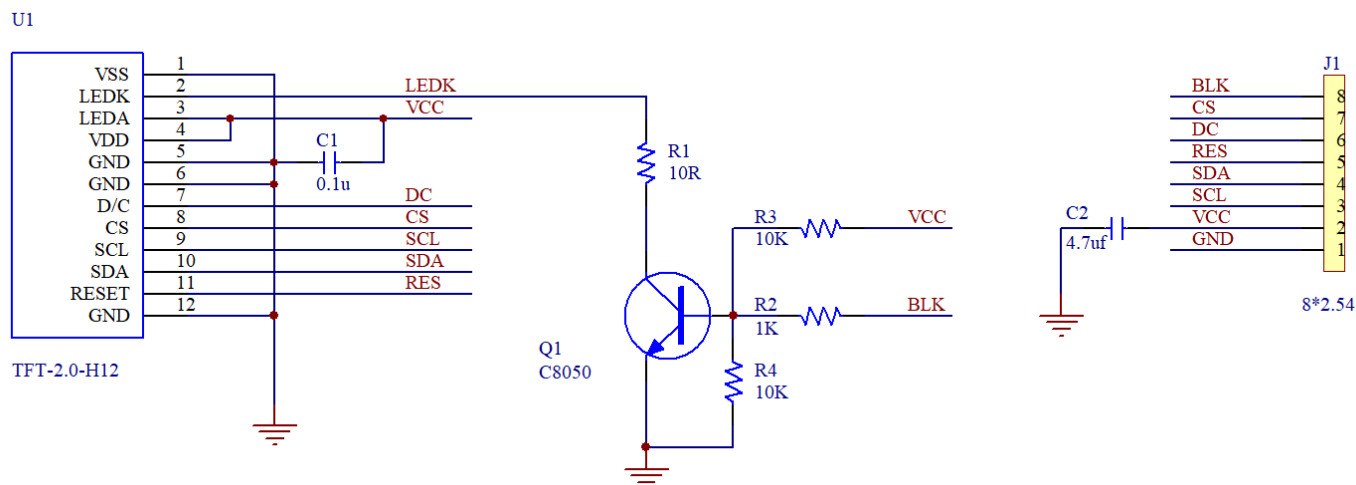
NOTES:

- 1.DISPLAY TYPE: 2.0" TFT
- 2.VIEWING DIRECTION: All
- 3.POLARIZER MODE: TRANSMISSIVE/NORMALLY BLACK
- 4.DRIVER IC: ST7789V2
- 5.Resolution: 240(H)RGB x320(V)
- 6.INTERFACE:4SPI
- 7.OPERATING TEMP: -20°C~70°C
- 8.STORAGE TEMP: -30°C~80°C
- 9.BACKLIGHT: 2 CHIP-WHITE LED
- 10.GENERAL TOLERANCE:±0.20
- 11.ROHS

Pin Descriptions

Pin	Symbol
1	GND
2	VCC
3	SCL
4	SDA
5	RES
6	DC
7	CS
8	BLK

7. Schematic diagram



8. Absolute Maximum Ratings

VSS=0V, Ta=25°C

Item		Symbol	Min.	Max.	Unit
Supply Voltage	Power supply	VCC	3.0	+3.6	V
	Analog	-	-	-	V
	IO	IOVCC	3.0	+3.6	V
Input Voltage		Vi	-0.3	IOVDD+0.3	V
Storage temperature		T stg	-30	+70	°C
Operating temperature		T op	-20	+60	°C
Storage humidity		H stg	10	Note 1	%RH
Operating humidity		H op	10	Note 1	%RH

Note 1: 90%RH max, If Ta is below 50°C; 60%RH max, If Ta is over 60°C.

9. Electrical Characteristics

DC Characteristics

Item		Symbol	Min.	Typ.	Max.	Unit
Supply Voltage	Power supply	VCC	3.0	3.3	3.6	V
Logic Low input voltage		VIL	-0.3IOVDD	-	0.3IOVDD	V
Logic High input voltage		VIH	0.7IOVDD	-	IOVDD	V
Logic Low output voltage		VOL	-	-	0.2IOVDD	V
Logic High output voltage		VOH	0.8IOVDD	-	-	V
Current Consumption	Normal display	Ivdd	-	50	-	mA
	Standby mode	Ivdd	-	20	-	uA
Frame Frequency		fFR	-	60	-	Hz

10 AC Characteristics

Reset timing and interface timing: Please refer to IC datasheet.

11 Command Table

Please refer to IC datasheet.

12 Recommended Setting and Initialization Flow for Reference

Please refer to attached file.

13 Optical Specifications

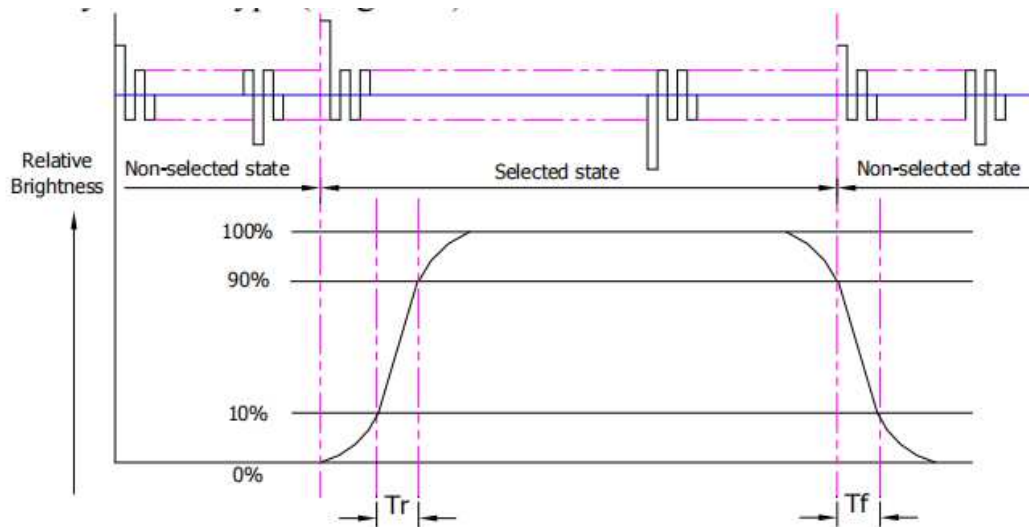
13.1 Optical Specifications

Ta=25°C, VDD=2.8V, TN LC+ Polarizer

Backlight On (Transmissive Mode)	Item		Symbol	Condition	Specification			Unit	
					Min.	Typ.	Max.		
	Luminance on surface(I _f =20mA)		Lv	Normally viewing angle θX = θY =0°	600	650	-	cd/m²	
	Contrast ratio		CR		-	600	-	-	
	Response time		TR	-	10	20	ms		
			TF	-	20	30			
	Chromaticity Transmissive		Red	XR	-	0.614	0.644	0.674	-
				YR		0.290	0.320	0.350	-
			Green	XG		0.270	0.300	0.330	-
				YG		0.540	0.570	0.600	-
			Blue	XB		0.104	0.134	0.164	-
				YB		0.097	0.127	0.157	-
			White	XW		0.267	0.297	0.327	-
				YW		0.302	0.332	0.362	-
	Viewing Angle		Horiz ontal	θX+	Center CR≥10	-	80	-	Deg.
				θX-		-	80	-	
Vertic al			θY+	-		80	-		
			θY-	-		80	-		
NTSC Ratio(Gamut)		-	-	-	60	-	%		

13.2 Definition of Response Time

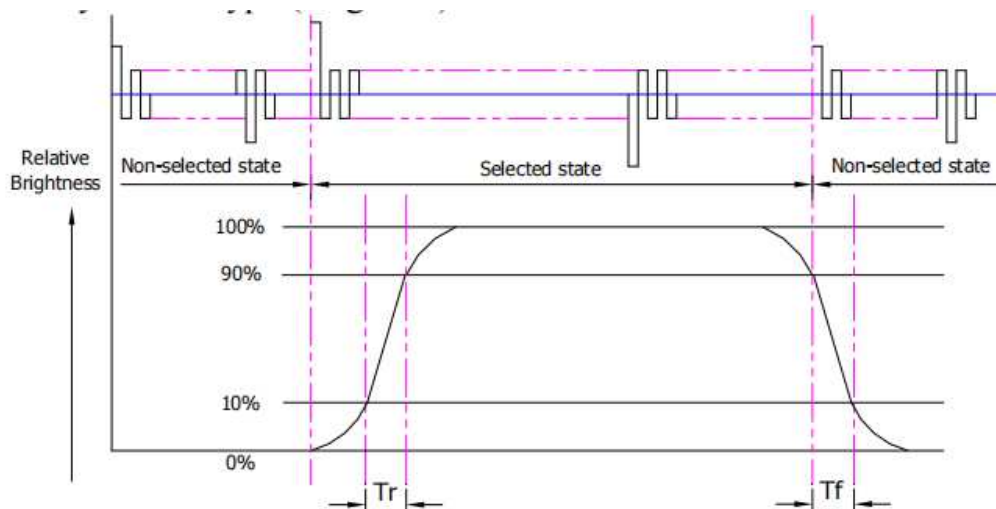
13.2.1 Normally Black Type (Negative)



T_r is the time it takes to change from non-selected state with relative luminance 10% to selected state with relative luminance 90%;

T_f is the time it takes to change from selected state with relative luminance 90% to non-selected state with relative luminance 10%.

13.2.2 Normally White Type (Positive)



T_r is the time it takes to change from non-selected state with relative luminance 90% to selected state with relative luminance 10%;

T_f is the time it takes to change from selected state with relative luminance 10% to non-selected state with relative luminance 90%;

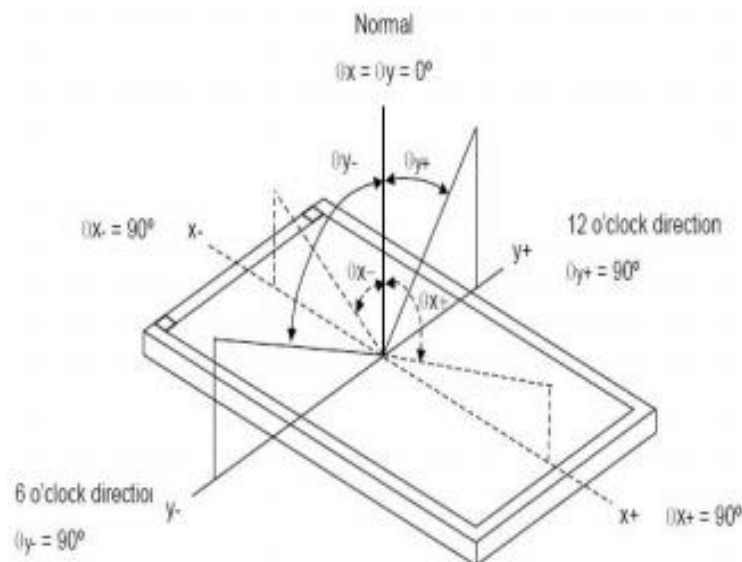
13.3 Definition of Contrast Ratio

Contrast is measured perpendicular to display surface in reflective and transmissive mode. The measurement condition is:

Measuring Equipment	BM-7 or EQUI
Measuring Point Diameter	3mm//1mm
Measuring Point Location	Active Area centre point
Test pattern	A: All Pixels white
	B: All Pixel black
Contrast setting	Maximum

Definitions: CR (Contrast) = Luminance ofWhite Pixel / Luminance ofBlack Pixel

13.4 Definition of Viewing Angles



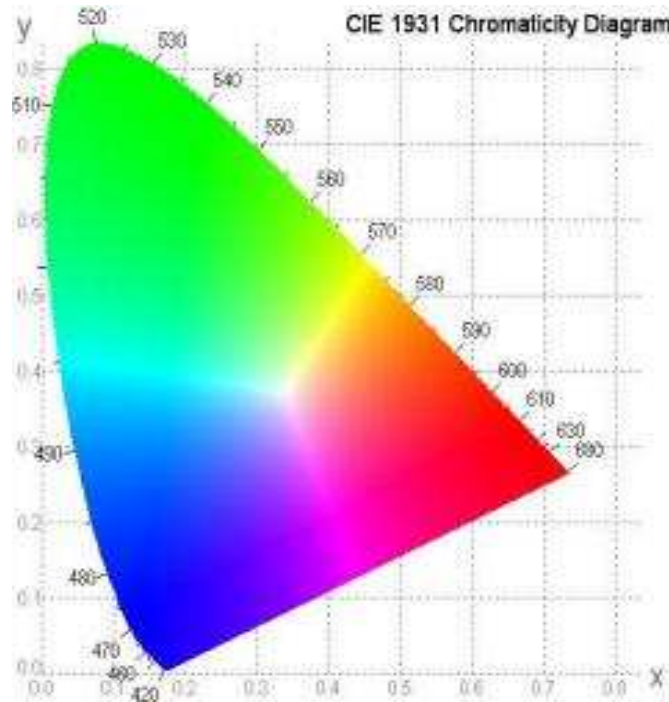
Measuring machine: LCD-5100 or EQUI

13.5 Definition of Color Appearance

R,G,B and W are defined by (x, y) on the TOPchromaticity diagram

NTSC=area ofRGB triangle/area ofNTSC triangleX100%

Measuring picture: Red, Green, Blue and White (Measuring machine: BM-7)



13.6 Definition of Surface Luminance, Uniformity and Transmittance

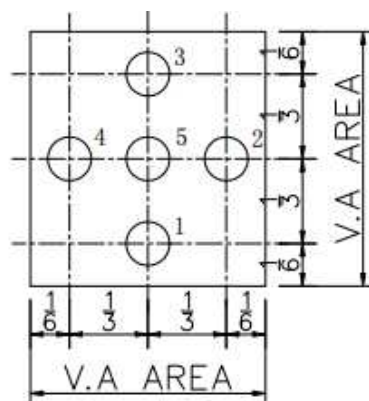
Using the transmissive mode measurement approach, measure the white screen luminance of the display panel and backlight.

13.6.1 Surface Luminance: $LV = \text{average (LP1:LP5)}$

13.6.2 Uniformity = $\text{Minimal (LP1:LP5)} / \text{Maximal (LP1:LP5)} * 100\%$

13.6.3 Transmittance = $LV \text{ on LCD} / LV \text{ on Backlight} * 100\%$

Note :Measuring machine:BM-7



14 Quality Assurance

14.1 Purpose

This standard for Quality Assurance assures the quality of LCD module products supplied to customer by lexcellence display.

14.2 Agreement Items

lexcellence and customer shall negotiate if the following situation occurs:

14.2.1 Discrepancies between lexcellence 's QA standards and customer's QA standards.

14.2.2 Additional requirement to be added in product specification.

14.2.3 Any other special problem.

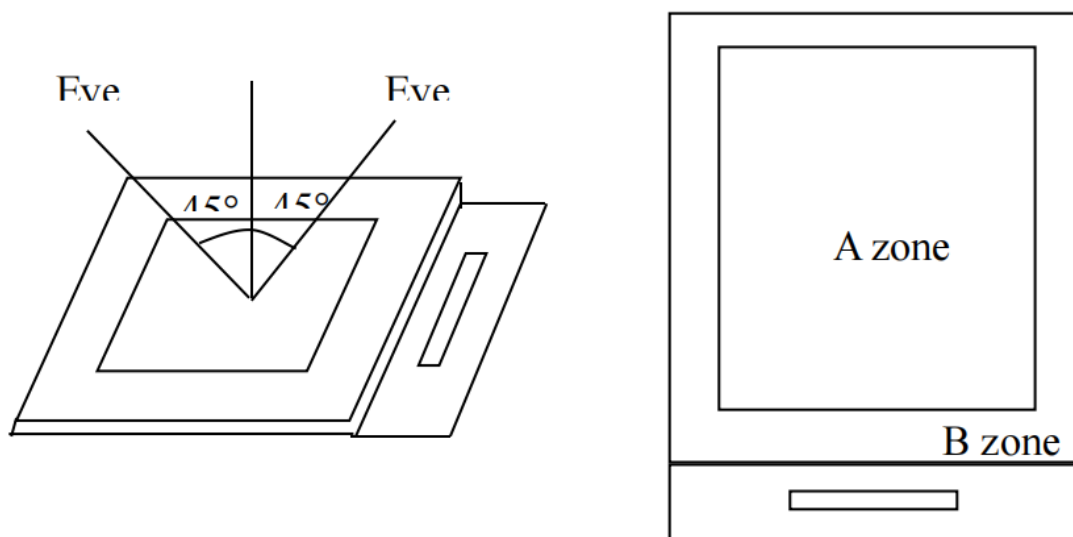
14.3 Standard of the Product Visual Inspection

14.3. 1 Appearance inspection:

14.3.1.1 The inspection must be under illumination about 1000 – 1500 lx, and the distance of view must be at 30cm $\pm$ 2cm.

14.3.1.2 The viewing angle should be 45° from the vertical line without reflection light or follows customer's viewing angle specifications.

14.3.1.3 Definition of area: A Zone: Active Area, B Zone: Viewing Area.



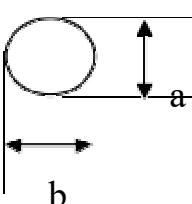
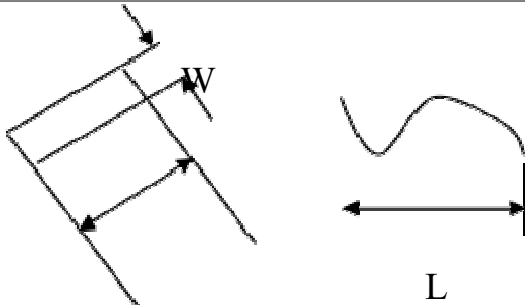
14.3.2 Basic principle: A set of sample to indicate the limit of acceptable quality level must be discussed by both lexcellence and customer when there is any dispute happened.

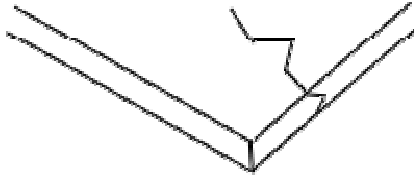
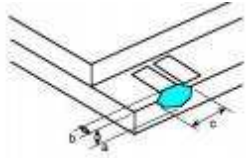
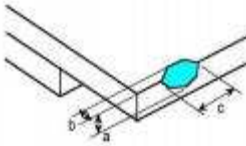
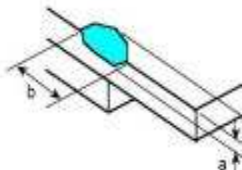
14.4 Inspection Specification

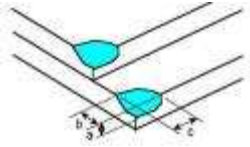
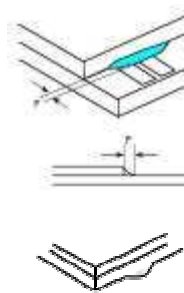
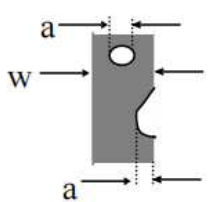
Sampling plan according to GB/T2828.1-2012/ISO 2859-1: 1999 and ANSI/ASQC Z1.4-1993, normal level 2 and based on:

Major defect: AQL 0.4

Minor defect: AQL 1.0

No.	Item	Criteria (Unit: mm)																			
01	Black / White spot Foreign material (Round type) Pinholes Stain Particles inside cell. (Minor defect)	 $\varphi = (a + b) / 2$	<table><tr><th>Size</th><th>Area</th><th>Acc. Qty</th></tr><tr><td>$\varphi \leq 0.10$</td><td></td><td>Ignore</td></tr><tr><td>$0.10 < \varphi \leq 0.15$</td><td></td><td>2</td></tr><tr><td>$0.15 < \varphi \leq 0.20$</td><td></td><td>1</td></tr><tr><td>$0.20 < \varphi$</td><td></td><td>0</td></tr><tr><td colspan="2">Total</td><td>2 (no include $\varphi \leq 0.10$)</td></tr></table>	Size	Area	Acc. Qty	$\varphi \leq 0.10$		Ignore	$0.10 < \varphi \leq 0.15$		2	$0.15 < \varphi \leq 0.20$		1	$0.20 < \varphi$		0	Total		2 (no include $\varphi \leq 0.10$)
		Size	Area	Acc. Qty																	
$\varphi \leq 0.10$		Ignore																			
$0.10 < \varphi \leq 0.15$		2																			
$0.15 < \varphi \leq 0.20$		1																			
$0.20 < \varphi$		0																			
Total		2 (no include $\varphi \leq 0.10$)																			
Distance between 2 defects should more than 5mm apart.																					
02	Black and White line Scratch Foreign material (Line type) (Minor defect)		<table><tr><th>Length</th><th>Width</th><th>Acc. Qty</th></tr><tr><td>/</td><td>$W \leq 0.03$</td><td>Ignore</td></tr><tr><td>$L \leq 2$</td><td>$0.03 < W \leq 0.05$</td><td>1</td></tr><tr><td>/</td><td>$0.05 < W$</td><td>0</td></tr><tr><td colspan="2">Total</td><td>1</td></tr></table>	Length	Width	Acc. Qty	/	$W \leq 0.03$	Ignore	$L \leq 2$	$0.03 < W \leq 0.05$	1	/	$0.05 < W$	0	Total		1			
		Length	Width	Acc. Qty																	
/	$W \leq 0.03$	Ignore																			
$L \leq 2$	$0.03 < W \leq 0.05$	1																			
/	$0.05 < W$	0																			
Total		1																			
Distance between 2 defects should more than 5mm apart. Scratches not viewable through the back of the display are acceptable.																					

No.	Item	Criteria (Unit: mm)										
03	Glass Crack (Minor defect)	 LCD with extensible crack line is unacceptable(When press the cracked LCD area, the line will expand, we define it is extensible crack line)										
04	Glass Chipping Pad Area: (Minor defect)	 <table><tr><th>Length and Width</th><th>Acc. Qty</th></tr><tr><td>$c < 5.0, b < 0.4$</td><td>Ignore</td></tr></table>	Length and Width	Acc. Qty	$c < 5.0, b < 0.4$	Ignore						
Length and Width	Acc. Qty											
$c < 5.0, b < 0.4$	Ignore											
05	Glass Chipping Rear of Pad Area: (Minor defect)	 <table><tr><th>Length and Width</th><th>Acc. Qty</th></tr><tr><td>$c > 3.0, b < 1.0$</td><td>1</td></tr><tr><td>$c < 3.0, b < 1.0$</td><td>2</td></tr><tr><td>$c < 3.0, b < 0.5$</td><td>4</td></tr><tr><td colspan="2">$a < \text{Glass Thickness}$</td></tr></table>	Length and Width	Acc. Qty	$c > 3.0, b < 1.0$	1	$c < 3.0, b < 1.0$	2	$c < 3.0, b < 0.5$	4	$a < \text{Glass Thickness}$	
Length and Width	Acc. Qty											
$c > 3.0, b < 1.0$	1											
$c < 3.0, b < 1.0$	2											
$c < 3.0, b < 0.5$	4											
$a < \text{Glass Thickness}$												
06	Glass Chipping Except Pad Area: (Minor defect)	 <table><tr><th>Length and Width</th><th>Acc. Qty</th></tr><tr><td>$c \leq 0.6, b < 5.0$</td><td>Ignore</td></tr><tr><td colspan="2">$a < \text{Glass Thickness}$</td></tr></table>	Length and Width	Acc. Qty	$c \leq 0.6, b < 5.0$	Ignore	$a < \text{Glass Thickness}$					
Length and Width	Acc. Qty											
$c \leq 0.6, b < 5.0$	Ignore											
$a < \text{Glass Thickness}$												

No.	Item	Criteria (Unit: mm)									
07	Glass Corner Chipping: (Minor defect) 	<table><tr><th>Length and Width</th><th>Acc. Qty</th></tr><tr><td>c < 2.0, b< 1.5</td><td>Ignore</td></tr><tr><td>c < 1.5, b< 2</td><td>Ignore</td></tr><tr><td colspan="2">a<Glass Thickness</td></tr></table>		Length and Width	Acc. Qty	c < 2.0, b< 1.5	Ignore	c < 1.5, b< 2	Ignore	a<Glass Thickness	
		Length and Width	Acc. Qty								
		c < 2.0, b< 1.5	Ignore								
		c < 1.5, b< 2	Ignore								
a<Glass Thickness											
08	Glass Burr: (Minor defect) 	Glass burr don't affect assemble and module dimension.									
		<table><tr><th>Length</th><th>Acc. Qty</th></tr><tr><td>F < 0.5</td><td>Ignore</td></tr></table>		Length	Acc. Qty	F < 0.5	Ignore				
		Length	Acc. Qty								
		F < 0.5	Ignore								
09	FPC Defect: (Minor defect) 	9.1 Dent, pinhole width a<w/2. (w: circuitry width.) 9.2 Open circuit is unacceptable. 9.3 No oxidation, contamination and distortion.									
10	Screen deformation 	Test for insertion of plug gauge at highest warping point: (0.96-3.1 inches does not contain 3.1) HVII0.25MM The client has special requirements, according to drawing									
11	Bubble on Polarizer (Minor defect)										
		Diameter	Acc. Qty								
		φ≤0.15	Ignore								
		0. 15 <φ≤0.20	2								
		0.20 <φ≤0.30	1								
		0.3 < φ	None								

No.	Item	Criteria (Unit: mm)	
12	Dent on Polarizer (Minor defect)	Diameter	Acc. Qty
		$\varphi \leq 0.15$	Ignore
		$0.15 < \varphi \leq 0.20$	2
		$0.20 < \varphi \leq 0.30$	1
		$0.3 < \varphi$	None
13	Bezel	13.1 No rust, distortion on the Bezel. 13.2 No visible fingerprints, stains or other contamination.	
14	Touch Panel	D: Diameter W: width L: length 14.1 Spot: $D \leq 0.20$ is acceptable $0.20 < D \leq 0.3$, acceptable QTY, 3 2dots are acceptable and the distance between defects should more than 5mm. $D > 0.3$ is unacceptable 14.2 Dent: $D > 0.30$ is unacceptable 14.3 Scratch: $W \leq 0.03$, $L \leq 10$ is acceptable, $0.03 < W \leq 0.10$, $L \leq 10$, acceptable QTY, 3 Distance between 2 defects should more than 5 mm. $W > 0.10$ is unacceptable.	
15	PCB	15.1 No distortion or contamination on PCB terminals. 15.2 All components on PCB must same as documented on the BOM/component layout. 15.3 Follow IPC-A-600F.	
16	Soldering	Follow IPC-A-610C standard	

No.	Item	Criteria (Unit: mm)
17	Electrical Defect (Major defect)	The below defects must be rejected. 17.1 Missing vertical / horizontal segment, 17.2 Abnormal Display. 17.3 No function or no display. 17.4 Current exceeds product specifications. 17.5 LCD viewing angle defect. 17.6 No Backlight. 17.7 Dark Backlight. 17.8 Touch Panel no function. 17.9 Dark Dot –one Allowed. 17. 10 Bright Dot – one Allowed. Remark: 1. A pixel defect is acceptable if one color is none functional and causes a bright dot. The display may have one case where one color is out and cause a dark dot. 2. Bright dot caused by scratch and foreign object accords to item1.
18	Leak	Yellow light,OK; White light,According to the limit sample

Remark: Visual and cosmetic defects are rejectable only if these fall within the LCD viewing area.

14.5 Classification of Defects

Visual defects (Except no / wrong label) are treated as minor defect and electrical defect is major.

14.6 Identification/marketing criteria

Any unit with illegible / wrong /double or no marking/ label shall be rejected.

14.7 Packing

14.7.1 There should be no damage ofthe outside carton box, each packaging box should has label in the correct location per packing drawing requirement.

14.7.2 All direct package materials shall offer ESD protection.

15 Reliability Specification

Item	Condition	Cycle Time	Quantity	Remark
Constant Temp. and Constant Humidity Operation Test	+40 ± 3°C, 90 ± 3%RH	96hrs	--	*1
High Temp. Operation Test	+70 ± 3°C	96hrs	--	
Low Temp. Operation Test	-20 ± 3°C	96hrs	--	
Thermal Shock Test	-20 ± 3°C (30min) +70 ± 3°C (30min)	10cycles	--	
ESD Test(end product)	150pF, 330Ω, ±2KV, Contact	10times	--	*2, *3
	150pF, 330Ω, ±6KV, Air			
Vibration Test (for packaging)	Frequency: 10Hz to 55Hz to 10Hz, Swing: 1.5mm, time : X, Y, Z each 2H.	6hrs	One inner carton	*4

Note 1. For humidity test, DI water should be used.

Inspection Standard: Inspect after 1-2hrs storage at room temperature, the sample shall be free from the following defects:

- I Air bubble in the LCD
- I Seal Leakage
- I Non-display
- I Missing Segment
- I Glass Crack
- I IDD is greater than twice initial value.
- I Others as per QA Inspection Criteria

Note 2. No defect is allowed after testing

The End Product ESD value is only indicative and depends on customer ESD protection design for the whole system.

Note 3. ESD should be applied to LCD glass panel, not other areas (such as on IC and so on) IDD should be within twice initial value.

In case of malfunction defect caused by ESD damage, if it would be recovered to normal state after resetting, it would be judged as a good part.

Note 4. Only upon request.

16 Precautions and Warranty

16.1 Safety

- 16.1.1 The liquid crystal in the LCD is poisonous. Do not put it in your mouth. If the liquid crystal touches your skin or clothes, wash it off immediately using soap and water.
- 16.1.2 Since the liquid crystal cells are made of glass, do not apply strong impact on them. Handle with care.

16.2 Handling

- 16.2.1 Reverse and use within ratings in order to keep performance and prevent damage.
- 16.2.2 Do not wipe the polarizer with dry cloth, as it might cause scratch. If the surface of the LCD needs to be cleaned, wipe it swiftly with cotton or other soft cloth soaked with petroleum IPA, do not use other chemicals.

16.3 Operation

- 16.3.1 Do not drive LCD with DC voltage
- 16.3.2 Response time will increase below lower temperature
- 16.3.3 Display may change color with different temperature
- 16.3.4 Mechanical disturbance during operation, such as pressing on the display area, may cause the segments to appear "fractured".

16.4 Static Electricity

- 16.4.1 CMOS LSIs are equipped in this unit, so care must be taken to avoid the electro-static charge, by ground human body, etc.
- 16.4.2 The normal static prevention measures should be observed for work clothes and benches.
- 16.4.3 The module should be kept into anti-static bags or other containers resistant to static for storage.

16.5 Limited Warranty

- 16.5.1 Unless otherwise agreed between KAC_DISPLAY and customer, KAC_DISPLAY will replace or repair any of its LCD and LCM which KAC_DISPLAY found to be defective electrically and visually when inspected in accordance with KAC_DISPLAY Quality Standards, for a period of

one year from date of shipment.

16.5.2 The warranty liability of KAC_DISPLAY is limited to repair and/or replacement. KAC_DISPLAY will not be responsible for any consequential loss.

16.5.3 If possible, we suggest you use up all modules in six months. If the module storage time over twelve months, we suggest that recheck it before the module be used.

17 Packaging

TBD

18 Prior Consult Matter

1. For KAC_DISPLAY standard products, we keep the right to change material, process for improving the product property without prior notice to our customer.
2. For OEM products, if any changes are needed which may affect the product property, we will consult with our customer in advance.
3. If you have special requirement about reliability condition, please let us know before you start the test on our samples.